

B_{BL} : A Bilateral Modal Logic for LLM Factuality Evaluation

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The Wrong Question

Prevailing approach to LLMs and logic

Can we get LLMs to reason logically?

Make the model do modus ponens, avoid contradictions, chain inferences.

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LLMs as defeasible derivability oracles

Sources that provide **provisional**, **non-monotone** verdicts about what is epistemically supportable — embedded in a logical calculus that makes those verdicts **compositionally tractable**.

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The locus of inference is the *framework*, not the model. The LLM answers bilateral verification queries; the logic does everything else.

Two Contributions

Contribution 1: $V_3 \times V_3$ is the right semantic space

Current LLM factuality evaluation treats verification as a single dimension — true, false, or uncertain (V_3). But **verifiability** and **refutability** are empirically independent. Collapsing them loses information: ignorance $\langle f, f \rangle$ and contradiction $\langle t, t \rangle$ are both “uncertain” in V_3 but have completely different epistemic profiles. $V_3 \times V_3$ preserves the distinction.

Contribution 2: $\Box p$ operationalises prompt-agnostic knowledge

Correct evaluation at a single phrasing s_0 may not reflect genuine knowledge. $\llbracket \Box p \rrbracket = \langle \min u_i, \max v_i \rangle$ formalises **reliable knowledge**: the bilateral truth value must be stable across semantic variations of the question. Prompt-sensitive knowledge (PSK) — $\llbracket p \rrbracket \in D$ but $\llbracket \Box p \rrbracket \notin D$ — is the formal safety-relevant gap.

B_{BL} : Object Language

Syntax. Propositional atoms $p, q, \dots$ closed under:

$$\varphi ::= p \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \vee \psi \mid \Box\varphi \mid \Diamond\varphi$$

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Bilateral truth value. $\langle u, v \rangle \in V_3 \times V_3$ where:

- ▶ $u =$ **verifiability**: degree to which p is supportable
- ▶ $v =$ **refutability**: degree to which $\neg p$ is supportable

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LLM-grounded valuation. $\llbracket p \rrbracket = \langle u, v \rangle$ via two *independent* queries:

$$u = \zeta^+(s_0, p) \qquad v = \zeta^-(s_0, p)$$

each by majority vote over $n = 3$ samples. The components are independent by design — the oracle is queried twice.

Propositional connectives

$$\neg \langle u, v \rangle = \langle v, u \rangle$$

$$\langle u_1, v_1 \rangle \wedge \langle u_2, v_2 \rangle = \langle \min(u_1, u_2), \max(v_1, v_2) \rangle$$

$$\langle u_1, v_1 \rangle \vee \langle u_2, v_2 \rangle = \langle \max(u_1, u_2), \min(v_1, v_2) \rangle$$

Negation swaps components:
refutability of p = verifiability of $\neg p$.

Modal operators over accessible situations $\{s_i\}$

$$\llbracket \Box p \rrbracket = \langle \min_i u_i, \max_i v_i \rangle$$

$$\llbracket \Diamond p \rrbracket = \langle \max_i u_i, \min_i v_i \rangle$$

Necessity is conservative:
weakest verification, strongest refutation across situations.

Truth Functions

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Compositional semantics for the oracle

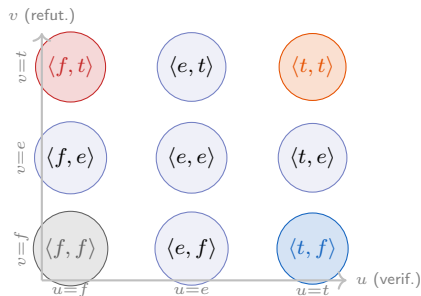
Conjunction propagates weakness: if either conjunct carries refutation evidence, the compound inherits it. $\llbracket \Box p \rrbracket$ filters out prompt-sensitive verdicts before they enter any inference chain.

The Bilattice NINE

$\text{NINE} = \langle V_3 \times V_3, \leq_t, \leq_k \rangle$ carries two independent lattice orderings:

$$\langle u_1, v_1 \rangle \leq_t \langle u_2, v_2 \rangle \Leftrightarrow u_1 \leq u_2 \text{ and } v_1 \geq v_2 \quad \langle u_1, v_1 \rangle \leq_k \langle u_2, v_2 \rangle \Leftrightarrow u_1 \leq u_2 \text{ and } v_1 \leq v_2$$

$\leq_t$: **truth** ordering $\leq_k$: **knowledge/information** ordering



Value	Semantics
$\langle t, f \rangle$	True: verified, not refuted
$\langle f, t \rangle$	False: refuted, not verified
$\langle f, f \rangle$	Ignorance: no signal
$\langle t, t \rangle$	Contradiction: conflicting
$\langle t, e \rangle, \langle e, f \rangle, \dots$	partial / asymmetric

$\langle f, f \rangle$ and $\langle t, t \rangle$ are both “uncertain” under unilateral evaluation. B_{BL} **makes the distinction** — and it is diagnostically consequential.

The LLM-Grounded Model $\mathcal{M}$

$$\mathcal{M} = \langle S, R_{\mathcal{M}}, V_{\mathcal{M}} \rangle$$

S — Situations

Natural-language descriptions: source questions, paraphrases, reorderings. $s_0 \in S$ is the **source situation** (the original question/claim context).

$R_{\mathcal{M}}$ — Accessibility

Meaning-preserving syntactic variation:

$s_0 R_{\mathcal{M}} s_i$ iff s_i is a paraphrase of s_0 .

Generated once per dataset via Claude Opus 4.6; cached and shared across all evaluation models.

$V_{\mathcal{M}}$ — Bilateral Valuation

$V_{\mathcal{M}}(s, p) = \langle u, v \rangle$ via two independent LLM queries:

$$u = \zeta^+(s, p) \text{ (verification)}$$

$$v = \zeta^-(s, p) \text{ (refutation)}$$

Each by majority vote over $n = 3$ calls to:

C^+ : *"Is p supported by evidence?"*

C^- : *"Is $\neg p$ supported by evidence?"*

The LLM is an oracle for $V_{\mathcal{M}}$. It is never asked to infer. All reasoning is done by the formal calculus over the $\langle u, v \rangle$ values it returns.

Epistemic Policies as B_{BLD,D^*} Parameterisation

The **projection operator** maps $\langle u, v \rangle$ to a classical verdict:

$$\pi_{D,D^*}(\langle u, v \rangle) = \begin{cases} \text{true} & \text{if } \langle u, v \rangle \in D \quad (\text{designated}) \\ \text{false} & \text{if } \langle u, v \rangle \in D^* \quad (\text{anti-designated}) \\ \text{abstain} & \text{otherwise (gap)} \end{cases}$$

Policy	D	D^*	Character
Classical	$\{\langle t, f \rangle\}$	$\{\langle f, t \rangle\}$	Only clean verified assertions designated
Paraconsistent	$\{\langle t, f \rangle, \langle t, t \rangle\}$	$\{\langle f, t \rangle, \langle e, t \rangle\}$	Tolerates contradictions; designates all verified
Paracomplete	$\{\langle t, f \rangle, \langle t, e \rangle\}$	$\{\langle f, t \rangle\}$	Tolerates gaps; designates partial verifications

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Key insight

Classical, paraconsistent, and paracomplete evaluation are not ad hoc thresholds — they are instances of B_{BLD,D^*} , differing *only* in the choice of designated values. The policy is a **logical parameter**, not an engineering choice.

Knowledge Stability: The Modal Operator $\Box$

Modal necessity aggregates $\llbracket p \rrbracket$ over s_0 and n paraphrase situations $\{s_1, \dots, s_n\}$:

$$\llbracket \Box p \rrbracket = \left\langle \min_t \{u_i\}, \max_t \{v_i\} \right\rangle \quad i = 0, 1, \dots, n$$

Epistemological reading:

$\llbracket \Box p \rrbracket \in D$ iff p is verifiable and not refuted across *all* syntactically accessible situations.

This implements the **safety condition** (Sosa/Williamson) and **tracking condition** (Nozick) from epistemology: knowledge requires that the verdict would not easily have been different in nearby situations.

Operationally:

1. Generate n paraphrases of s_0
2. Evaluate $\llbracket p \rrbracket$ at each s_i
3. Aggregate: take min verifiability, max refutability

Necessity is *conservative*: any single situation that refutes p pulls $\llbracket \Box p \rrbracket$ away from $\langle t, f \rangle$.

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Modal possibility $\Diamond$

$\llbracket \Diamond p \rrbracket = \langle \max_t \{u_i\}, \min_t \{v_i\} \rangle$ — p is supportable from *at least one* accessible situation. The B axiom $p \rightarrow \Box \Diamond p$ connects the two: stable truth implies witnessed possibility.

Prompt-Sensitive Knowledge (PSK)

Definition

p is **prompt-sensitive knowledge** at s_0 if

$$\llbracket p \rrbracket \in D \quad \text{but} \quad \llbracket \Box p \rrbracket \notin D$$

The model *appears* to know p at s_0 but the verdict does not survive syntactic variation.

Why PSK is not genuine knowledge:

- ▶ **Fails safety** (Sosa/Williamson): the model would have produced a different verdict in a nearby accessible situation
- ▶ **Fails tracking** (Nozick): the mechanism is sensitive to surface form, not semantic content
- ▶ **Fails reliabilism**: the process is not truth-conducive across the relevant reference class of inputs

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PSK is diagnostic

PSK signals *where* the model's knowledge is fragile — a target for fine-tuning, retrieval augmentation, or explicit uncertainty.

$\llbracket \Box p \rrbracket$ is the filter

Inference chains should be built on $\llbracket \Box p \rrbracket \in D$, not $\llbracket p \rrbracket \in D$. Modal stability is the right epistemic threshold for reasoning.

How ζ^+ and ζ^- Are Computed

Verification query ζ^+ (forced-choice):

You are tasked with determining whether the following assertion can be verified as true based on available evidence and knowledge.

Assertion: *The answer to “In Severance, whom does Mark Scout replace as department head?” is Peter “Petey” Kilmer.*

Context: *In Severance, whom does Mark Scout replace as department head?*

You must respond with exactly one of: VERIFIED
CANNOT VERIFY

Response:

VERIFIED	$\rightarrow u = t$
CANNOT VERIFY	$\rightarrow u = f$
<i>parse failure</i>	$\rightarrow u = e$

Issued $n = 3$ times each; majority vote $\Rightarrow \llbracket p \rrbracket = \langle u, v \rangle$. Parse failure (e) = model could not commit. Components are **entirely independent**: the model never sees both questions.

Refutation query ζ^- (forced-choice):

You are tasked with determining whether the following assertion can be refuted (shown to be false) based on available evidence and knowledge.

Assertion: *The answer to “In Severance, whom does Mark Scout replace as department head?” is Peter “Petey” Kilmer.*

Context: *In Severance, whom does Mark Scout replace as department head?*

You must respond with exactly one of: REFUTED
CANNOT REFUTE

Response:

REFUTED	$\rightarrow v = t$
CANNOT REFUTE	$\rightarrow v = f$
<i>parse failure</i>	$\rightarrow v = e$

Assertion (ground truth: correct)

*The answer to “Who was the lawyer who developed matrix algebra and also worked on higher-dimensional geometry?” is **Arthur Cayley**.*

Sit.	Paraphrase	$\llbracket p \rrbracket$
s_0	<i>Who was the lawyer who developed matrix algebra...?</i>	$\langle t, f \rangle$
s_1	<i>Which lawyer is known to have developed matrix algebra...?</i>	$\langle t, f \rangle$
s_2	<i>Who was the attorney who developed matrix algebra...?</i>	$\langle t, f \rangle$
s_3	<i>Can you identify the lawyer responsible for matrix algebra...?</i>	$\langle t, f \rangle$

$$\begin{aligned}\llbracket \Box p \rrbracket &= \langle \min(t, t, t, t), \\ &\quad \max(f, f, f, f) \rangle \\ &= \langle t, f \rangle\end{aligned}$$

$$\llbracket p \rrbracket \in D \text{ and } \llbracket \Box p \rrbracket \in D$$

Stable knowledge. The verdict $\langle t, f \rangle$ is robust across all four situations — not prompt-sensitive.

Assertion (ground truth: correct)

The answer to “In Severance, whom does Mark Scout replace as department head?” is Peter “Petey” Kilmer.

Sit.	Paraphrase	$\llbracket p \rrbracket$
s_0	<i>In Severance, whom does Mark Scout replace as dept. head?</i>	$\langle t, f \rangle \checkmark$
s_1	<i>In Severance, who is the person that Mark Scout succeeds?</i>	$\langle f, f \rangle \times$
s_2	<i>Whom does Mark Scout take over from in the role of dept. head?</i>	$\langle t, f \rangle \checkmark$
s_3	<i>Which individual did Mark Scout replace as dept. head?</i>	$\langle t, f \rangle \checkmark$

$$\begin{aligned} \llbracket \Box p \rrbracket &= \langle \min(t, f, t, t), \\ &\quad \max(f, f, f, f) \rangle \\ &= \langle f, f \rangle \end{aligned}$$

$$\llbracket p \rrbracket \in D \text{ but } \llbracket \Box p \rrbracket \notin D$$

Prompt-sensitive knowledge. s_1 (“succeeds” for “replace”) collapses verdict to $\langle f, f \rangle$. Fails safety.

Assertion (ground truth: **incorrect**)

*The answer to “What was the full name of the first Prime Minister of Congo?” is **Beyoncé**.*

Sit.	Paraphrase	$\llbracket p \rrbracket$
s_0	What was the full name of the first Prime Minister of Congo?	$\langle f, t \rangle \checkmark$
s_1	What was the complete name of Congo's first Prime Minister?	$\langle f, t \rangle \checkmark$
s_2	What was the full name of the individual who served as first PM?	$\langle f, t \rangle \checkmark$
s_3	What was the first Prime Minister of Congo's full name?	$\langle f, t \rangle \checkmark$

$$\begin{aligned}\llbracket \Box p \rrbracket &= \langle \min(f, f, f, f), \\ &\quad \max(t, t, t, t) \rangle \\ &= \langle f, t \rangle\end{aligned}$$

$$\llbracket p \rrbracket \in D^* \quad \textbf{and} \quad \llbracket \Box p \rrbracket \in D^*$$

Stable refutation. The model robustly identifies this absurd claim as false — syntactically invariant anti-knowledge.

Assertion (ground truth: **incorrect**)

The answer to “In the British series Happy Valley Season 2, who jumps off a bridge to their death?” is Richard Wadsworth.

Sit.	Paraphrase	$\llbracket p \rrbracket$
s_0	<i>In Happy Valley S2, who jumps off a bridge to their death?</i>	$\langle t, t \rangle$
s_1	<i>Which character jumps off a bridge to their death in Happy Valley S2?</i>	$\langle t, t \rangle$
s_2	<i>Who is the character that leaps from a bridge in Happy Valley S2?</i>	$\langle t, t \rangle$
s_3	<i>Which character dies by jumping from a bridge in Happy Valley S2?</i>	$\langle t, t \rangle$

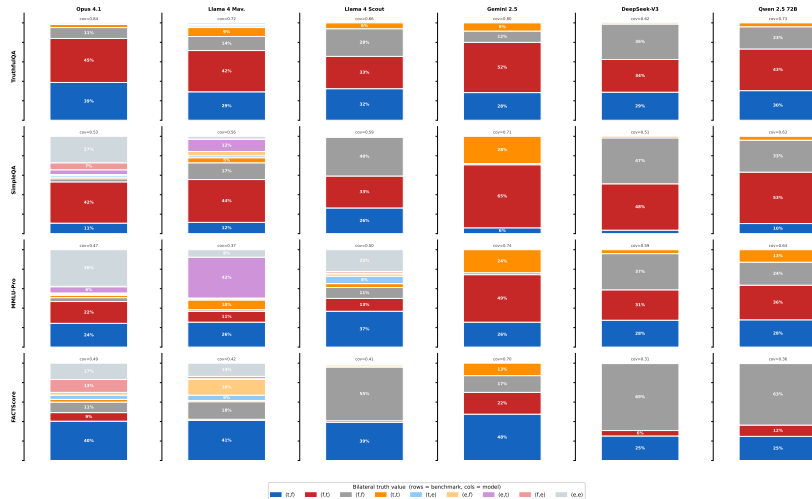
$$\begin{aligned} \llbracket \Box p \rrbracket &= \langle \min(t, t, t, t), \\ &\quad \max(t, t, t, t) \rangle \\ &= \langle t, t \rangle \end{aligned}$$

$$\llbracket p \rrbracket \notin D \cup D^*$$

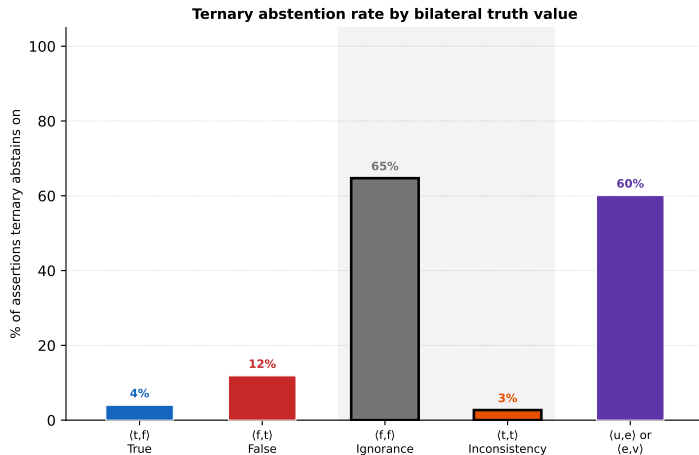
Stable contradiction. The model both verifies and refutes the claim across all four situations. This is representational conflict — not ignorance. Unilateral evaluation returns “uncertain”; B_{BL} identifies **CONTRADICTION**.

Bilateral TV Distributions: $[p]$

Bilateral Truth-Value Distributions — VM(ss, p) (N=250, N=3 majority vote per call)
 % within each cell sums to 100. 'cov' = fraction predicted under classical epistemic policy.

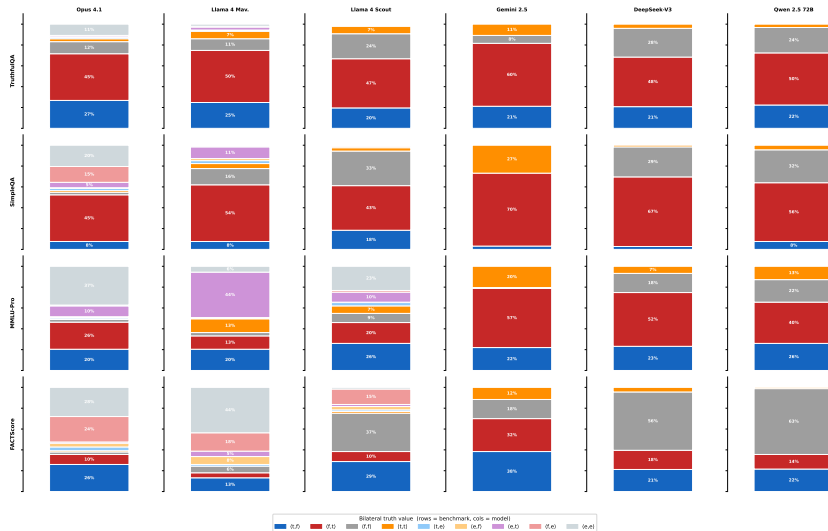


Ignorance $\langle f, f \rangle$ and Contradiction $\langle t, t \rangle$ Are Not the Same



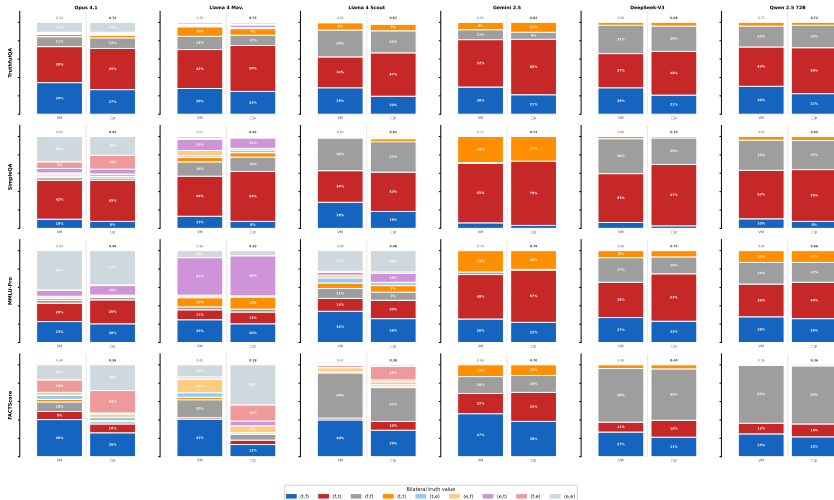
Both $\langle f, f \rangle$ and $\langle t, t \rangle$ fall outside $D \cup D^*$ under every epistemic policy — both are “*uncertain*” to any unilateral approach. **Bilateral evaluation separates them.** $\langle f, f \rangle$: near-random ground-truth positive rate (52%), ternary correctly abstains 65% of the time. $\langle t, t \rangle$: ground-truth positive rate skews 68%, ternary almost never abstains (3%) — it commits on the verification signal while missing the refutation evidence entirely.

Modal TV Distributions: $\llbracket \square p \rrbracket$



$\llbracket p \rrbracket$ vs $\llbracket \Box p \rrbracket$: Modal Tightening

VM(s_0, p) vs $\llbracket \Box p \rrbracket$ Bilateral TV Distributions
Left bar = source situation only; Right bar = modal necessity over $s_0 + 2$ paraphrases. Numbers above bars = classical coverage.



Empirical Setup

Models ($\times 6$, open-source-skewed):

Model	Provider
Claude Opus 4.1	Anthropic
Llama 4 Maverick	Meta
Llama 4 Scout	Meta
Gemini 2.5 Flash	Google
DeepSeek-V3	DeepSeek
Qwen 2.5 72B	Alibaba

Benchmarks ($\times 4$):

Benchmark	Focus
TruthfulQA	Misconception resistance
SimpleQA	Factual precision
MMLU-Pro	Expert knowledge
FACTScore	Biographical facts

Protocol: $N = 250$ assertions per cell (seed = 42, balanced 50/50 +/-); $n = 3$ paraphrase variants; bilateral samples = 3 (majority vote). Variants generated once per dataset, cached across all models.

Five approaches compared in a single pass: bilateral (3 policies), forced unilateral, ternary, and confidence@0.5.

Bilateral vs. Ternary vs. Unilateral: F1 Macro and Coverage

Macro-averages across 6 models per benchmark ($N = 250$ each). Bilateral uses the paracomplete policy $D = \{\langle t, f \rangle, \langle t, e \rangle\}$. **Bold** = highest F1 in that row.

Benchmark	Bilateral (PC)		Ternary		Unilateral	
	F1	Cov	F1	Cov	F1	Cov
TruthfulQA	.74	.94	.84	.84	.78	1.00
SimpleQA	.79	.85	.87	.72	.73	1.00
FACTScore	.45	.90	.51	.62	.60	1.00
Average	.66	.89	.74	.73	.70	1.00

Ternary achieves the highest F1 on average — by refusing to answer 27% of assertions. Unilateral commits on all assertions but cannot express what it does not know. **Neither can distinguish $\langle f, f \rangle$ (ignorance) from $\langle t, t \rangle$ (contradiction)**: the ternary abstain-rate analysis (previous slide) shows why that distinction matters.

Coverage and F1 Macro: Source vs. Modal

Each cell: **source** / **modal** with delta (Δ). Coverage = fraction of assertions predicted under classical $D = \{\langle t, f \rangle\}$. F1 macro computed on covered assertions only.

Model	TruthfulQA		SimpleQA		MMLU-Pro		FACTScore	
	Cov	F1	Cov	F1	Cov	F1	Cov	F1
	src / modal (Δ modal—src)							
Opus 4.1	.74/.72 (-.02)	.91/.86 (-.05)	.52/.52 (.00)	.93/.87 (-.06)	.43/.46 (+.04)	.90/.84 (-.06)	.49/.36 (-.14)	.52/.59 (+.07)
Llama 4 Mav.	.70/.75 (+.04)	.77/.74 (-.03)	.57/.62 (+.05)	.86/.73 (-.13)	.36/.33 (-.02)	.79/.80 (+.01)	.42/.18 (-.24)	.43/.50 (+.07)
Llama 4 Scout	.62/.67 (+.04)	.72/.63 (-.10)	.63/.61 (-.02)	.91/.88 (-.03)	.48/.46 (-.02)	.74/.77 (+.03)	.42/.38 (-.04)	.40/.56 (+.16)
Gemini 2.5 F.	.81/.82 (+.01)	.83/.76 (-.06)	.71/.73 (+.02)	.62/.52 (-.10)	.74/.79 (+.05)	.78/.75 (-.03)	.69/.70 (+.01)	.64/.68 (+.04)
DeepSeek-V3	.66/.68 (+.02)	.81/.75 (-.06)	.60/.70 (+.10)	.72/.52 (-.20)	.66/.75 (+.09)	.78/.76 (-.02)	.38/.40 (+.02)	.57/.58 (+.01)
Qwen 2.5 72B	.73/.72 (-.01)	.83/.78 (-.05)	.63/.64 (+.01)	.77/.73 (-.05)	.64/.66 (+.02)	.76/.75 (-.01)	.36/.36 (-.01)	.57/.55 (-.01)

Red: large PSK-driven coverage loss (Llama 4 Mav. -0.24 on FACTScore).

Blue: F1 rises under modal filtering — prompt-sensitive lucky predictions are correctly withheld.

Empirical Results: Bilateral Distributions

Key findings across 24 model $\times$ benchmark cells:

- ▶ $\langle t, f \rangle$ (TRUE) dominates **TruthfulQA** — models are mostly confident and uncontested on factual questions
- ▶ $\langle f, f \rangle$ (IGNORANCE) dominates **FACTScore** — models systematically cannot verify *or* refute biographical facts
- ▶ **MMLU-Pro** shows lowest classical coverage — questions require genuine uncertainty; paraconsistent policy recovers significant F1
- ▶ $\langle f, f \rangle$ and $\langle t, t \rangle$ are both “uncertain” under unilateral evaluation but are observably *different* states with different diagnostic implications

The diagnostic reading

High $\langle f, f \rangle$ on FACTScore is not a failure of bilateral evaluation — it is the framework **correctly diagnosing** that the model has no relevant training signal for biographical claims.

Unilateral evaluation would report the same result as MMLU-Pro: low accuracy. Bilateral evaluation distinguishes *why*.

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- ▶ $\langle f, f \rangle$ (IGNORANCE) dominates **FACTScore** — models systematically cannot verify *or* refute biographical facts
- ▶ **MMLU-Pro** shows lowest classical coverage — questions require genuine uncertainty; paraconsistent policy recovers significant F1
- ▶ $\langle f, f \rangle$ and $\langle t, t \rangle$ are both “uncertain” under unilateral evaluation but are observably *different* states with different diagnostic implications

Bilateral $\approx$ ternary on covered examples. The contribution is not accuracy improvement — it is the richer epistemic characterisation of the non-covered cases.

The diagnostic reading

High $\langle f, f \rangle$ on FACTScore is not a failure of bilateral evaluation — it is the framework **correctly diagnosing** that the model has no relevant training signal for biographical claims.

Unilateral evaluation would report the same result as MMLU-Pro: low accuracy. Bilateral evaluation distinguishes *why*.

Frame Conditions: Empirical Support for System B

T axiom: $\llbracket \Box p \rrbracket \leq_t \llbracket p \rrbracket$ Confirmed

Necessity can only *weaken* the source verdict. If $\llbracket p \rrbracket = \langle t, f \rangle$ then $\llbracket \Box p \rrbracket \in \{\langle t, f \rangle, \langle t, e \rangle, \langle e, f \rangle, \dots\}$ — never strictly more designated.

Tested on 277 cases where removing s_0 from the aggregation would change the result: **0 violations** across all 6 models $\times$ 4 benchmarks.

B axiom: $\llbracket p \rrbracket \in D \Rightarrow \llbracket \Box p \rrbracket_{s_i} \in D$

Consistent

If p holds at s_0 , it should hold modally from any variant too. Proxy test: recompute $\llbracket \Box p \rrbracket$ treating each s_i as the new source in place of s_0 . Does the designation result agree?

95.7% agreement (6,300 model $\times$ question pairs, each tested 3–4 $\times$ across variants).

Theoretical grounding: paraphrase accessibility is symmetric by construction.

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Completeness: the Kamide–Shramko translation

B_{BL} is sound and complete w.r.t. system B frames via translation τ to classical modal logic. **Def. 19** ($\tau_D(\varphi) = \varphi_v \wedge \varphi_r^1$) is the *only* policy-sensitive part — the rest is frame-condition agnostic. Metatheoretic results (decidability, complexity) are inherited from classical modal T/B.

Modal modus ponens

$$\llbracket \Box p \rrbracket \in D \wedge \llbracket \Box(p \rightarrow q) \rrbracket \in D \implies \llbracket \Box q \rrbracket \in D$$

The LLM evaluates $\llbracket \Box p \rrbracket$ and $\llbracket \Box(p \rightarrow q) \rrbracket$ independently. The logic derives $\llbracket \Box q \rrbracket$. **The model never performs the inference** — only the oracle queries.

In compound AI systems:

- ▶ **Multi-hop reasoning:** bilateral values propagate through inference chains; errors are formally tracked, not silently absorbed
- ▶ **RAG pipelines:** $\langle t, t \rangle$ surfaces conflicting retrieved vs. prior evidence rather than masking it
- ▶ **Multi-model aggregation:** conjunction/disjunction of bilateral values is formal, not a voting heuristic

Why current pipelines lack this:

- ▶ Chain-of-thought asks the LLM to *perform* the inference — unaccountable and unverifiable
- ▶ Confidence scores conflate ignorance $\langle f, f \rangle$ and contradiction $\langle t, t \rangle$
- ▶ No formal criterion for when to trust a composed result

Formal Reasoning with Defeasible Derivability Oracles

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B_{BL} provides the **formal infrastructure** that makes LLM outputs usable as inputs to rigorous inference. Not: make LLMs reason logically. Rather: **reason logically using LLMs**.

Open Questions

1. **Neighbourhood semantics** — non-normal modal operators for LLMs that do not respect the distribution axiom K; more principled basis for the accessibility structure
2. **Systematic D -parameterisation study** — how does (D, D^*) affect coverage, F1, and calibration? LP- B_{BL} and K3- B_{BL} as proof-theoretic targets
3. **Multi-agent B_{BL}** — different LLMs instantiating different accessibility relations $R_{\mathcal{M}_1}, R_{\mathcal{M}_2}$; formal aggregation across models
4. **Dynamic operators** — belief revision as modal transition; fine-tuning as a map $\mathcal{M} \rightarrow \mathcal{M}'$ with formally characterised effects on $V_{\mathcal{M}}$
5. **Definitive symmetry test** — generate variants-of-variants to empirically confirm the B axiom, or establish system T + theoretical symmetry as the right frame class

Conclusion

The thesis

Don't try to get LLMs to reason logically. Build a logic that reasons using LLMs as defeasible derivability oracles.

B_{BL} delivers:

- ▶ A **sound and complete** bilateral modal logic grounded in LLM evaluation
- ▶ A **parameterised family** B_{BLD,D^*} of epistemic policies with a unified proof-theoretic treatment
- ▶ A **modal operationalisation** of knowledge stability via $\llbracket \Box p \rrbracket$, diagnosing prompt-sensitive knowledge before it enters inference chains
- ▶ **Empirically confirmed** frame conditions: T axiom (277/277), B axiom consistent (95.7% permuted agreement)
- ▶ A **formal basis** for reasoning about composed LLM outputs in compound AI systems

`arxiv.org/abs/2507.09751` · Code & data: `github.com/bradleyallen/bilateral-truth`